Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching

Project code: PKV Group: GFA26SE111

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I. Record of Changes

Date	A* M, D	In charge	Change Description
27/10/2026	A	Nguyễn Gia Huy	v0.1 – Test formula, workflows and demo scenarios prepared for the Review 2 commitment (BM04).
06/11/2026	A	Nguyễn Văn Thành Đạt	v0.2 – First test documentation: 116 planned cases, 36 unit checks, no results.
11/11/2026	M	Nguyễn Gia Huy	v1.0 – Rewritten: scope with levels, entry and exit criteria; test types with objective, technique and completion criteria; tools with vendor and version; dated test milestones; 114 concrete test cases mapped to FR-01–FR-33; real results of 09–11/11/2026 (515 EditMode tests passed; 49 passed, 3 failed, 62 pending system cases); 18 defects; analysis against the release criteria.
11/11/2026	M	Nguyễn Gia Huy	v1.1 – Human resources and workflow test cases: owners aligned with BM04.
12/11/2026	M	Nguyễn Gia Huy	v1.2 – Revised after the second internal review: EditMode run of 516 tests on main @ 5963501 (EV-06) and Play Mode check of fixed defects (EV-07); defect statuses with fix commits; new defects D-19 and D-20; quick doubles case TC-F07-04; workflow names as in BM04; CI and Android status stated; counts in the text computed from the case list.
13/11/2026	M	Nguyễn Gia Huy	v1.3 – Updated after the fixes of 12–13/11/2026 (main @ d9200cc): the GitHub Actions pipeline passes (EV-08), so 14 cases are now passed by the IntegrationChecks of its validate stage; transport and secret check EV-09 (D-18 fixed, D-17 partly fixed, TC-N09 still failed); Android APK built (EV-10) and defect D-21 found and fixed; D-20 accepted as designed; CI history corrected.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

The target of test is PickleVerse release 1.0 as specified in Report 3: the ten features FE-01 to FE-10 with their functional requirements FR-01 to FR-33, the business rules BR-01 to BR-51, and the non-functional requirements NFR-01 to NFR-12. Testing covers the Unity game client (Windows build and

Unity Editor), the dedicated game server, the Web API with its workers, the AI coaching service and the administration web application deployed on the staging server.

In scope. All screens and non-screen functions of Report 3, the gameplay rules and physics parameters of Report 4 §3.2, online matchmaking and dedicated play, data integrity of rewards and results, role-based access, and the measurable quality attributes NFR-01 to NFR-12.

Out of scope. iOS builds and real-money payment (Report 1, LI-02 and LI-06); load beyond the API capacity target, because the staging deployment has one dedicated-server slot (LI-07); third-party services themselves (Photon Cloud, Gmail SMTP), which are only tested through their integration points; and the tier model's accuracy until enough expert-labelled data exists (NFR-06).

Table 1. Test levels: responsibility, inputs, focus and exit criteria

Level	In charge	Inputs and time	Focus	Acceptance (exit) criteria
Unit	Each module owner; review by Trần Nguyễn Hiếu Thuận	Source code, BR-xx; before every merge	Rules, scoring, Elo, input decoding, economy, offline store, matchmaking policies (Unity EditMode, pytest)	All unit tests pass; every public rule method has at least one normal, abnormal and boundary case
Integration	Nguyễn Văn Thành Đạt	API build, Database.sql seed, PostgreSQL and Redis; each package milestone	API contracts, database writes, dedicated tickets and leases, training sync receipts, role and scope checks	All IntegrationChecks pass on a fresh database; no contract difference between Unity, API and AI service
System	Trần Nguyễn Hiếu Thuận with the feature owners	Staging deployment, test accounts and seed data; Rounds 1–3 (§3.3)	Every FR through the user interface, state transitions, NFR measurements	≥ 90% of system cases passed, all FRs covered by a passed case, no open Critical/High defect
Acceptance	Huỳnh Trung Kiên with the supervisor	Release candidate, demo scenarios W01–W04	End-to-end workflows of the four members	All four workflows pass in front of the supervisor

Constraints and assumptions.

- The staging server runs one dedicated-server container, so online cases that need two simultaneous matches cannot be executed.
- The GitHub Actions pipeline passes since 13/11/2026 (EV-08) and runs the IntegrationChecks, the FastAPI tests and the admin web type check for the parts that a push changes, but it has no Unity test step. The Unity EditMode suite is therefore still run with the Unity CLI on a team PC and the result file is stored with the commit hash; its last run is on main @ 5963501, before the client address changed in be556a1.
- An Android APK was built on 12/11/2026 (EV-10), but no Android phone has been used for testing yet. That APK predates the HTTPS switch, so the Android cases (TC-N03, TC-N10) need a new APK from main @ d9200cc and a reference device in Round 2.
- Tests that need several players (rooms, friends, reports) require two or more accounts and devices at the same time.
- The admin web moved to https://tad-min.io.vn on 13/11/2026 and was unreachable between the deployments of 01:18 and 14:34 that day; the admin web cases (TC-F15 to TC-F18, TC-W04) are run against the new address in Round 2.
- A dedicated test mailbox and seed accounts (for example a player at level 24 or with 9 passed lessons) must be added to Database.sql before Round 2.
- The Photon free plan limits concurrent users; group playtests are scheduled so that the limit is not reached.

2. Test Strategy

2.1 Testing Types

Table 2. Testing types

Type	Objective	Technique	Completion criteria
Functional	Verify every FR and BR	Black-box: equivalence partitions and boundary values from BR-xx; decision tables for scoring and Elo	Each FR has ≥ 1 passed case; all Main cases passed
Gameplay rules and physics	Verify serve, faults, scoring, contact timing and bounce	Automated EditMode tests on the engine-independent rule classes; scripted rallies	All rule tests pass; every fault type has a test
User interface and usability	Verify layout, messages and learnability	Checklist per screen against Report 3 §3.1 and the message list; moderated usability test (NFR-01)	No overlapping or truncated elements; NFR-01 met
Integration / API contract	Verify API, database and service contracts	IntegrationChecks console harness on a fresh database; shared JSON fixtures in contracts/	All checks pass on the release commit
Security	Verify authentication, authorisation and secret handling	Negative API tests with wrong roles and scopes, token expiry, rate limit; secret scan	NFR-09 checklist satisfied; no secret in the repository
Performance and load	Verify NFR-02 to NFR-05	Unity Profiler on reference devices; Fusion RTT statistics; k6 load test on the API	All measured values within the NFR limits
Network resilience	Verify reconnection and interruption	Disconnect a client for 20 s and 60 s; drop the dedicated server heartbeat	Behaviour matches BR-28 and BR-29
Compatibility	Verify target platforms	Install and smoke test on Windows 10/11 and Android 8+ (ARM64); admin web on Chrome and Edge	NFR-10 met
AI quality	Verify coaching findings and the gated model	Holdout evaluation against expert labels	NFR-06 thresholds met, otherwise the model stays disabled
Exploratory playtest and balance	Find defects that scripted cases miss; tune bot difficulty	Time-boxed sessions with a charter; win rate of players of known level against each bot level	Each session logged; balance targets agreed with the supervisor
Regression	Keep fixed defects fixed	Re-run the automated suites and the cases linked to closed defects in every round	No reopened defect

2.2 Test Levels

Table 3. Test types per level

Type of Tests	Unit	Integration	System	Acceptance
Functional	X	X	X	X
Gameplay rules and physics	X		X	X
User interface and usability			X	X
Integration / API contract		X		
Security	X	X	X	
Performance and load		X	X	
Network resilience	X		X	
Compatibility			X	X
AI quality	X		X	
Exploratory playtest and balance			X	X
Regression	X	X	X	

2.3 Supporting Tools

Table 4. Supporting tools

Purpose	Tool	Vendor/In-house	Version
Unity unit tests (EditMode, NUnit)	Unity Test Framework	Unity Technologies	1.8.0

Purpose	Tool	Vendor/In-house	Version
Run tests and capture screens from the command line	Unity CLI with the Pipeline package	Unity Technologies	CLI 1.0.0-beta.17, Pipeline 0.9.0-exp.1
Python unit tests	pytest (with httpx)	Open source	8.x
API and database integration checks	IntegrationChecks console harness	In-house	main @ d9200cc (run by the CI pipeline)
Manual API calls	Swagger UI	Swashbuckle (open source)	10.3.1
API load test (NFR-04)	k6	Grafana Labs (open source)	latest stable, planned
Frame-rate measurement	Unity Profiler	Unity Technologies	6000.6.2f1
Network statistics	Photon Fusion statistics and dashboard	Exit Games	Fusion 2.1.3
Test case and result management	Microsoft Excel (Report5 workbooks)	Microsoft	Microsoft 365
Continuous integration	GitHub Actions (plan, checks, build, publish and deploy jobs)	GitHub	Hosted runner ubuntu-24.04; runs 27, 28 and 30 of 13/11/2026 passed every job

3. Test Plan

3.1 Human Resources

Table 5. Human resources

Worker/Doer	Role	Specific Responsibilities/Comments
Trần Nguyễn Hiếu Thuận	Test lead	Owns the test plan and workbooks; runs system rounds; tracks defects; tests the account, onboarding, offline and social screens (F01, F02, F12, F14); owner of workflow W01
Nguyễn Gia Huy	Tester, gameplay owner	Unit tests of rules, physics, training and bot; playtests and balance sessions; runs the Unity EditMode suite; F03, F05, F11, F13; owner of workflow W02, support for W01 and W03
Nguyễn Văn Thành Đạt	Tester, backend and network owner	IntegrationChecks, API security and load tests, online and dedicated cases (F04, F06–F10); owner of workflow W03, support for W02
Huỳnh Trung Kiên	Tester, leader	Admin web cases (F15–F18), acceptance session with the supervisor, release decision; owner of workflow W04
Mr. Lại Đức Hùng	Supervisor	Witnesses the acceptance scenarios and approves the release criteria

3.2 Test Environment

Table 6. Test environment

Purpose	Tool	Provider	Version
Staging server	Ubuntu with Docker Engine and Docker Compose (API, AI service, dedicated server, PostgreSQL, Redis, S3 storage, web) behind a Caddy reverse proxy; API and admin web on https://tad-min.io.vn since 13/11/2026	Team VPS tad-min.io.vn	Ubuntu 24.04; images published by the CI run of main
Game client under test	Unity Editor and Windows build	Unity Technologies	6000.6.2f1
Test PC	Windows PC with keyboard and mouse	Team member	Windows 11 Home
Android device	Android phone (ARM64) for NFR-03 and NFR-10; APK of EV-10	Team member	Android 8.0 or later (to be recorded in Round 2)
Real-time service	Photon Cloud	Exit Games	Fusion 2.1.3
Database and cache	PostgreSQL, Redis	Docker images postgres:16-alpine and redis:7.4-alpine	16 / 7.4
Admin web browser	Google Chrome, Microsoft Edge	Google, Microsoft	Latest stable

3.3 Test Milestones

Table 7. Test milestones

Milestone Task	Start Date	End Date
Round 1 – automated suites and first system pass on staging	09/11/2026	13/11/2026
Prepare seed accounts, test mailbox and analytics seed	16/11/2026	18/11/2026
Round 2 – all system cases; retest of the fixed defects and fix verification of D-01 to D-21	16/11/2026	27/11/2026
Android install, smoke and frame-rate test on a reference phone (NFR-03, NFR-10)	16/11/2026	27/11/2026
Usability test with 10 students (NFR-01)	23/11/2026	27/11/2026
Load, latency and frame-rate measurements (NFR-02 to NFR-05)	23/11/2026	29/11/2026
Uptime probe for NFR-07 (14 days)	30/11/2026	13/12/2026
Round 3 – regression on the release candidate	30/11/2026	06/12/2026
Acceptance test W01–W04 with the supervisor	07/12/2026	11/12/2026

4. Test Cases

Unit test cases are documented in **Report5_Unit Test.xlsx** as decision tables (conditions, confirmations, type N/A/B and result for each UTCID) for ten core functions, each mapped to the NUnit test that implements it. Integration, system and acceptance test cases are documented in **Report5_Test Report.xlsx**, one sheet per feature group, with ID, description, procedure with concrete data, expected result, pre-conditions, three test rounds and a note that names the evidence or the reason why a case is pending.

The number of cases follows the Review 2 formula: 18 committed features × 3 cases (54) + 4 workflows + 42 state transitions (three per state machine SM-01 to SM-14) + 12 NFRs = 112 cases. Six cases are added: two for the expert review tool (FE-10), which is not in the Review 2 feature list, three for FR-03 to FR-05, which the Review 2 feature F01 did not detail, and one for quick doubles (TC-F07-04), which F07 describes only in general, giving **118 cases**.

Table 8. Test cases by sheet and level

Sheet	Scope	Cases	Unit	Integration	System	Acceptance	Automated
FE01 Account	FE-01 Account and onboarding	6	1	0	5	0	1
FE02 Training	FE-02 Tutorial, training and offline	9	4	0	5	0	4
FE03 Bot	FE-03 Bot practice	3	2	0	1	0	2
FE04 Online	FE-04 Online matches	13	8	0	5	0	8
FE05 Coaching	FE-05 AI coaching	3	1	0	2	0	1
FE06 Progress	FE-06 Progress, rewards and shop	6	4	1	1	0	5
FE07 Social	FE-07 History, social and ranking	6	2	1	3	0	2
FE08-09 Admin	FE-08/FE-09 Moderation and administration	12	1	2	9	0	2
FE10 Expert	FE-10 Expert review	2	0	1	1	0	1
Workflows	End-to-end workflows W01–W04	4	0	0	0	4	0
States	State transitions SM-01 to SM-14	42	11	27	4	0	22
NFR	Non-functional requirements	12	0	0	12	0	0
Total		**118**	**34**	**32**	**48**	**4**	**48**

4.1 Requirement Traceability

Table 9. Traceability from functional requirements to test cases

Requirement	Test cases	Cases	Passed	Coverage status
FR-01	TC-F01-01, TC-F01-03, TC-W01	3	1	Covered
FR-02	TC-F01-01, TC-F01-02	2	1	Covered

Requirement	Test cases	Cases	Passed	Coverage status
FR-03	TC-F01-04	1	0	Not yet covered
FR-04	TC-F01-05	1	0	Not yet covered
FR-05	TC-F01-06	1	1	Covered
FR-06	TC-F02-01, TC-F02-02, TC-W01	3	1	Covered
FR-07	TC-F03-01, TC-F03-03	2	2	Covered
FR-08	TC-F03-01, TC-F03-02, TC-W01	3	2	Covered
FR-09	TC-F14-01, TC-F14-02, TC-F14-03, TC-F06-03, TC-W01	5	2	Covered
FR-10	TC-F11-01	1	1	Covered
FR-11	TC-F02-03	1	1	Covered
FR-12	TC-F05-01, TC-F05-03, TC-F09-02	3	3	Covered
FR-13	TC-F05-01, TC-F05-02, TC-F05-03, TC-W02	4	4	Covered
FR-14	TC-F07-01, TC-F07-02, TC-F07-03, TC-F07-04, TC-F10-02, TC-W03	6	4	Covered
FR-15	TC-F08-01, TC-F08-02, TC-F08-03	3	2	Covered
FR-16	TC-F09-01, TC-F09-02, TC-W03	3	2	Covered
FR-17	TC-F09-03	1	0	Not yet covered
FR-18	TC-F09-01, TC-F10-01, TC-F10-02, TC-F10-03, TC-W02, TC-W03	6	5	Covered
FR-19	TC-F06-01, TC-F06-02, TC-F06-03, TC-W02	4	3	Covered
FR-20	TC-F04-03	1	1	Covered
FR-21	TC-F13-01, TC-F13-02, TC-F13-03	3	3	Covered
FR-22	TC-F04-01	1	1	Covered
FR-23	TC-F04-02	1	1	Covered
FR-24	TC-F11-02, TC-F11-03	2	1	Covered
FR-25	TC-F12-01, TC-F12-02	2	1	Covered
FR-26	TC-F12-03	1	1	Covered
FR-27	TC-F17-01, TC-W04	2	0	Not yet covered
FR-28	TC-F17-03, TC-W04	2	0	Not yet covered
FR-29	TC-F15-01, TC-F15-02, TC-F15-03	3	0	Not yet covered
FR-30	TC-F17-01, TC-W04	2	0	Not yet covered
FR-31	TC-F16-01, TC-F16-02, TC-F16-03, TC-W04	4	1	Covered
FR-32	TC-F17-02, TC-F18-01, TC-F18-02, TC-F18-03	4	1	Covered
FR-33	TC-X01-01, TC-X01-02	2	1	Covered

26 of the 33 functional requirements are covered by at least one passed case after Round 1 and the CI run of 13/11/2026 (EV-08); FR-03, FR-04, FR-17, FR-27, FR-28, FR-29, FR-30 have no passed case yet. NFR-01 to NFR-12 are traced one-to-one to TC-N01 to TC-N12: 1 of the 12 have been executed (TC-N09 failed), none has passed.

5. Test Reports

5.1 Evidence

Table 10. Test evidence

ID	Run	Date	Build / commit	Result	Location
EV-01	Unity EditMode suite through the Unity CLI (Pipeline test run)	11/11/2026	main @ 7f8ee32	515 tests, 515 passed, 0 failed, 6.19 s	Doc/Project_report/Report/evidence/editmode_results_20261111.json
EV-02	Playtest on the staging server (offline training, signed-in menu, bot match, AI coaching, ranked 1v1 with bot fallback)	10/11/2026	main @ 82f6a7a client; staging images of 06/11/2026	See Report5_Test Report.xlsx	screenshots/01–28, evidence/playtest_log_20261110.md
EV-03	UI check of the register, verification, password-reset screens and the tutorial	11/11/2026	main @ 7f8ee32	See TC-F01-03, TC-F02-01	screenshots/29–34
EV-04	Match history of the team test accounts (two ranked 1v1 attempts)	09/11/2026	staging	Both interrupted	screenshots/12_history.png

ID	Run	Date	Build / commit	Result	Location
EV-05	Team-recorded IntegrationChecks on disposable PostgreSQL databases (12 areas)	06/11/2026	df5f671	Reported as passed; logs not in the repository, so not counted	Review3_Evidence.md (file names and SHA-256)
EV-06	Unity EditMode suite through the Unity CLI after the UI and camera fixes (adds the camera framing test)	11/11/2026	main @ 5963501	516 tests, 516 passed, 0 failed, 5.96 s	Doc/Project_report/Report/evidence/editmode_results_20261111_5963501.json
EV-07	Unity Play Mode check of the fixed defects D-03, D-04, D-05, D-07 and D-19 (menu, mode selection, bot setup, bot match, training HUD)	11/11/2026	main @ 5963501	All five behave as expected; to be retested on staging in Round 2	screenshots/05, 11, 16, 17, 19
EV-08	GitHub Actions run 30 of the workflow "Build, publish and deploy PickleVerse": Backend integration checks (12 IntegrationChecks groups on PostgreSQL 16 and Redis 7.4), FastAPI tests, admin web type check, Unity Linux dedicated-server build, publish with smoke test, deployment over SSH. Runs 27 and 28 passed on main @ 1dfefe9	13/11/2026 14:20–14:34	main @ d9200cc	All nine jobs and all their steps succeeded (14 min)	evidence/ci_runs_20261113.json; github.com/Tad-Min/PickleVerse/actions/runs/43385071385
EV-09	Transport and secret check: HTTPS, redirects and open ports of the staging host; credential values in the tracked settings files. Run at 11:30 and again at 16:21 after commit be556a1 was deployed	13/11/2026	staging; main @ 1dfefe9 and d9200cc	16:21: API and admin web only over HTTPS, port 8080 closed, settings files clean; old values remain in the Git history	evidence/staging_check_20261113.md
EV-10	Android build with Unity batch mode and inspection of the APK manifest (aapt2, apksigner)	12/11/2026	main @ 5963501	APK built (190.5 MB, IL2CPP, ARM64, API 26–36); not installed on a device; cleartext HTTP forbidden (D-21, fixed in be556a1)	evidence/android_build_20261112.md

5.2 Unit Test Results

Table 11. Unity EditMode results by assembly (EV-06)

Test assembly	Tests	Passed	Failed
Audio.Tests	23	23	0
FrontEnd.Tests	4	4	0
HybridBot.Tests	15	15	0
LocalMatch.Tests	157	157	0
Multiplayer.Tests	101	101	0
Progression.Tests	125	125	0
Replays.Tests	16	16	0
Services.Tests	31	31	0
Shop.Tests	33	33	0
Social.Tests	11	11	0
Total	**516**	**516**	**0**

The table above shows the latest run (EV-06, 516 tests); the Round 1 run EV-01 on main @ 7f8ee32 had 515 tests, all passed, and EV-06 adds the camera framing regression test of D-03. The ten decision tables of Report5_Unit Test.xlsx contain 66 unit test cases (normal, abnormal and boundary), all passed in EV-01 and EV-06. The other automated suites run in the GitHub Actions pipeline. In run 30 on main @ d9200cc (EV-08) the jobs "Backend integration checks" (twelve IntegrationChecks groups: client-security, api-contract, schema, status-enums, content, training-sync, progress-reset, moderation, analytics, expert-review, dedicated and hybrid, each on a disposable PostgreSQL 16

database with Redis 7.4), "FastAPI tests" (pytest of the AI service) and "Admin web type check" all succeeded. A check group stops with an error at its first failed assertion, so a successful step means that every assertion passed. The backend integration checks have passed in every run since run 20 of 11/11/2026 (twelve runs); the FastAPI tests failed once, in run 29, and passed again in run 30. The earlier team runs of 06/11/2026 (EV-05) stay uncounted because their logs are not in the repository. The Unity EditMode suite is not part of the pipeline yet, and the job logs are kept by GitHub, not in the repository; the step results are stored in evidence/ci_runs_20261113.json. Code coverage has not been measured yet; the Unity Code Coverage package will be enabled in Round 2.

5.3 System and Integration Test Results

Table 12. Results by module after Round 1 and the CI run of 13/11/2026

Module	Cases	Passed	Failed	Pending
FE-01 Account and onboarding	6	3	0	3
FE-02 Tutorial, training and offline	9	7	0	2
FE-03 Bot practice	3	3	0	0
FE-04 Online matches	13	8	2	3
FE-05 AI coaching	3	2	1	0
FE-06 Progress, rewards and shop	6	6	0	0
FE-07 History, social and ranking	6	4	0	2
FE-08/FE-09 Moderation and administration	12	2	0	10
FE-10 Expert review	2	1	0	1
End-to-end workflows W01–W04	4	2	0	2
State transitions SM-01 to SM-14	42	26	0	16
Non-functional requirements	12	0	1	11
Total	**118**	**64**	**4**	**50**

Table 13. Release criteria on 13/11/2026

Metric	Value	Target	Status
Test coverage (executed / total)	57.6% (68 / 118)	100% before release	Not met
Pass rate of executed cases	94.1% (64 / 68)	≥ 90%	Met for executed cases
Test successful coverage (passed / total)	54.2%	≥ 90%	Not met
Automated cases	40.7% (48 / 118)	≥ 50%, run in CI	Not met (14 of them run in CI)
Functional requirements covered by a passed case	26 / 33	33 / 33	Not met
Open Critical / High defects	0 / 2 (D-09, D-17)	0 / 0	Not met
Non-functional requirements measured and met	1 measured, 0 met	12 / 12	Not met

5.4 Defects

Table 14. Defects found in Round 1 and in the checks of 11–13/11/2026

ID	Summary	Type	Severity	Found by	Owner	Status
D-01	Offline banner overlaps the titles of Progress, the training HUD and the lesson list	UI	Low	TC-F14-01	Thuận	Open
D-02	Offline: the analysis area stays at "Đang phân tích…" forever	Functional	Medium	TC-F06-03	Đạt	Open
D-03	Key-hint box (shot cue) covers the player character in training, bot and online matches	UI	Low	TC-F03-01	Huy	Fixed (5963501)
D-04	"Quyết định & Khiếu nại" button overlaps the PLAY button and the profile panel	UI	Medium	TC-F01-02	Thuận	Fixed (31a7c2e)
D-05	Fourth bot difficulty shows the technical name "BhTree"	UI text	Low	TC-F05-01	Huy	Fixed (d2fc5be)
D-06	Bot card says "Không cần kết nối mạng" but bot matches are disabled offline	UI text	Low	TC-F14-01	Thuận	Open

ID	Summary	Type	Severity	Found by	Owner	Status
D-07	Male/female choice is not applied in bot match, training and tutorial (always the female character)	Functional	Medium	TC-F05-01	Huy	Fixed (6d1ed37)
D-08	New players start at Elo 100 while the admin reset-Elo default is 2000	Business rule	Medium	TC-F10-01	Đạt	Open
D-09	Ranked 1v1 between two humans ended "Gián đoạn" twice, without replay	Functional	High	TC-F09-01	Đạt	Open
D-10	"Xem phân tích" button on the match result is oversized and covers the centre of the screen	UI	Low	TC-F05-01	Thuận	Open
D-11	Coaching popup mixes session-wide (2/86) and per-technique (1/16) figures	Usability	Low	TC-F06-01	Đạt	Open
D-12	Elite Token has no earning source, so GEMSTONE can only be granted at level 25	Requirement	Medium	TC-F04-01	Kiên	Open
D-13	Hard bot is beaten 2–0 by the auto-player with short rallies; balance not validated	Balance	Medium	TC-F05-01	Huy	Open
D-14	After a normal ranked result the lobby shows "Mất kết nối Photon."	UI message	Low	TC-W03	Đạt	Open
D-15	Ranked bot offered about 160 s after the search started instead of 180 s	Functional	Low	TC-F07-01	Đạt	Open
D-16	Sync-UnityTests.ps1 fails to parse in Windows PowerShell 5.1 (UTF-8 file without BOM)	Tooling	Medium	Unit test run	Thuận	Open
D-17	Development credentials committed in appsettings.json and a fallback JWT key in Program.cs	Security	High	TC-N09	Đạt	Partly fixed
D-18	Staging API and admin web are served over HTTP instead of HTTPS	Security	High	TC-N09	Đạt	Fixed (be556a1)
D-19	Settings tile is pushed below the screen after the Decisions and Appeals tile was added to the menu	UI	Low	TC-F01-02	Huy	Fixed (9dd994c)
D-20	Level 25 needs 60,000 XP in total (n × 200 per level) while a match gives 5–10 XP, so the GEMSTONE paddle (BR-37) is effectively unreachable	Requirement	Medium	TC-F04-03	Kiên	Accepted
D-21	The Android APK declares usesCleartextTraffic=false (manifest written by the App UI package) while BackendApiSettings used http://tad-min.io.vn:8080, so API calls from an Android phone are expected to be refused	Configuration	High	TC-N10	Huy	Fixed (be556a1)

Round 1 (09–11/11/2026) and the follow-up checks of 11–13/11/2026 found 21 defects: 4 High, 8 Medium and 9 Low. 7 of them (D-03, D-04, D-05, D-07, D-18, D-19, D-21) are fixed in code; the fix commit is shown in the Status column. The five user-interface fixes were checked in Unity Play Mode (EV-07), the HTTPS switch of D-18 was verified on staging (EV-09), and D-21 is fixed by the same commit but not yet verified with a new Android build. Fixed defects stay in the Fixed state until the Round 2 retest. 13 defects still need work, including 2 High defects (D-09, D-17).

- **D-17 – Partly fixed (13/11/2026).** Commit be556a1 emptied the credential values in appsettings.json and removed the fallback JWT key from Program.cs (the API now refuses to start without a key of at least 32 bytes). The old values are still readable in the Git history, so they must be rotated on staging and the rotation confirmed.
- **D-20 – Accepted (13/11/2026).** Team decision of 13/11/2026: accepted as designed, the XP curve is not changed; closed without a fix.

Severity follows Report 3 NFR-08: Critical means loss of player data or inability to sign in or play; High means a main workflow or a security requirement fails; Medium means a requirement is partly wrong or a rule is inconsistent; Low means a cosmetic or wording problem. The defect life cycle (Open → Fixed → Verified → Closed) is tracked in PickleVerse_Project_Tracking.xlsx.

5.5 Analysis and Next Actions

- **Engine-level quality is good.** All 516 EditMode tests pass (EV-06), and every system case that could be executed for the single-player features (training, bot match, coaching, progress screens) passed, apart from the offline analysis status (D-02).

- **Online play between humans is the main risk.** The only human-versus-human ranked attempts (EV-04) were interrupted (D-09), which also blocks the reconnection, report and appeal cases. Reproducing D-09 with server logs is the first task of Round 2 (owner: Đạt, by 18/11/2026).
- **Security improved on 13/11/2026 but is not closed.** Commit be556a1 moved the game client, the API and the admin web to https://tad-min.io.vn, closed port 8080 and removed the credentials from the settings file and the fallback JWT key from the code; EV-09 confirms the transport part (D-18 fixed). The removed values are still readable in the Git history, so D-17 stays partly fixed and TC-N09 failed until the team rotates them on staging and records the rotation; the rest of the NFR-09 checklist and a secret scan in the pipeline are planned for Round 2 (owner: Đạt).
- **Coverage grew through the pipeline, and is now limited by test data.** The CI run EV-08 moved 14 cases to Passed and three more requirements (FR-20, FR-32, FR-33) to Covered. 50 cases are still pending, mainly because seed accounts, a test mailbox and a second player were not available; 16 of them are partly covered by a CI check and still need the missing step named in their note. The seed data will be added to Database.sql before Round 2 (owner: Thuận, by 18/11/2026).
- **Automation is below the 50% target.** 48 of 118 cases are automated: 34 by the Unity EditMode suite, which is still run by hand, and 14 by the IntegrationChecks of the CI validate job. Next: add the EditMode suite as a GameCI unity-test-runner step that reuses the licence secrets of the dedicated-server build, and add the check groups that the job does not run yet (admin authorisation, shared replays, social, password recovery) so that the remaining state-transition cases can be linked to them.
- **7 defects are fixed but not yet closed.** D-03, D-04, D-05, D-07 and D-19 were fixed on 11/11/2026 and checked in Play Mode; D-18 and D-21 were fixed by be556a1 on 13/11/2026. All are closed only after the Round 2 retest on the staging build. D-20 (60,000 XP for the GEMSTONE level) was accepted as designed on 13/11/2026 and is closed without a fix; D-12 (no Elite Token source) is still open.
- **Android is built but untested.** An APK was built from main @ 5963501 on 12/11/2026 (EV-10) and its inspection found D-21; no Android device has run the game yet. A new APK from main @ d9200cc, which uses the HTTPS address, is needed for TC-N03 and TC-N10 in Round 2.
- **The delivery pipeline works.** Of the 32 runs, 29 did not pass: two configuration failures on 05/11, 18 runs that did not start while the GitHub account was locked for a billing issue from 06/11 to 11/11, six dedicated-server build failures, one failed FastAPI test and two failed deployment steps (one of them a manual re-run on 13/11 at 16:07). Runs 27, 28 and 30 of 13/11/2026 completed checks, build, publication and deployment in about 15 minutes each, and run 30 deployed the release that is on staging now.